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PCRpal

Python CLI tool for batch validation and analysis of PCR primers

GIT



MOTIVATION

PCR primer design is one of the most common steps in molecular biology workflows. Existing tools (Primer3, OligoCalc) require manual input of sequences one by one. PCRpal allows batch validation of multiple primers at once from a single input file, saving time and reducing human error.

HOW DOES IT WORK?

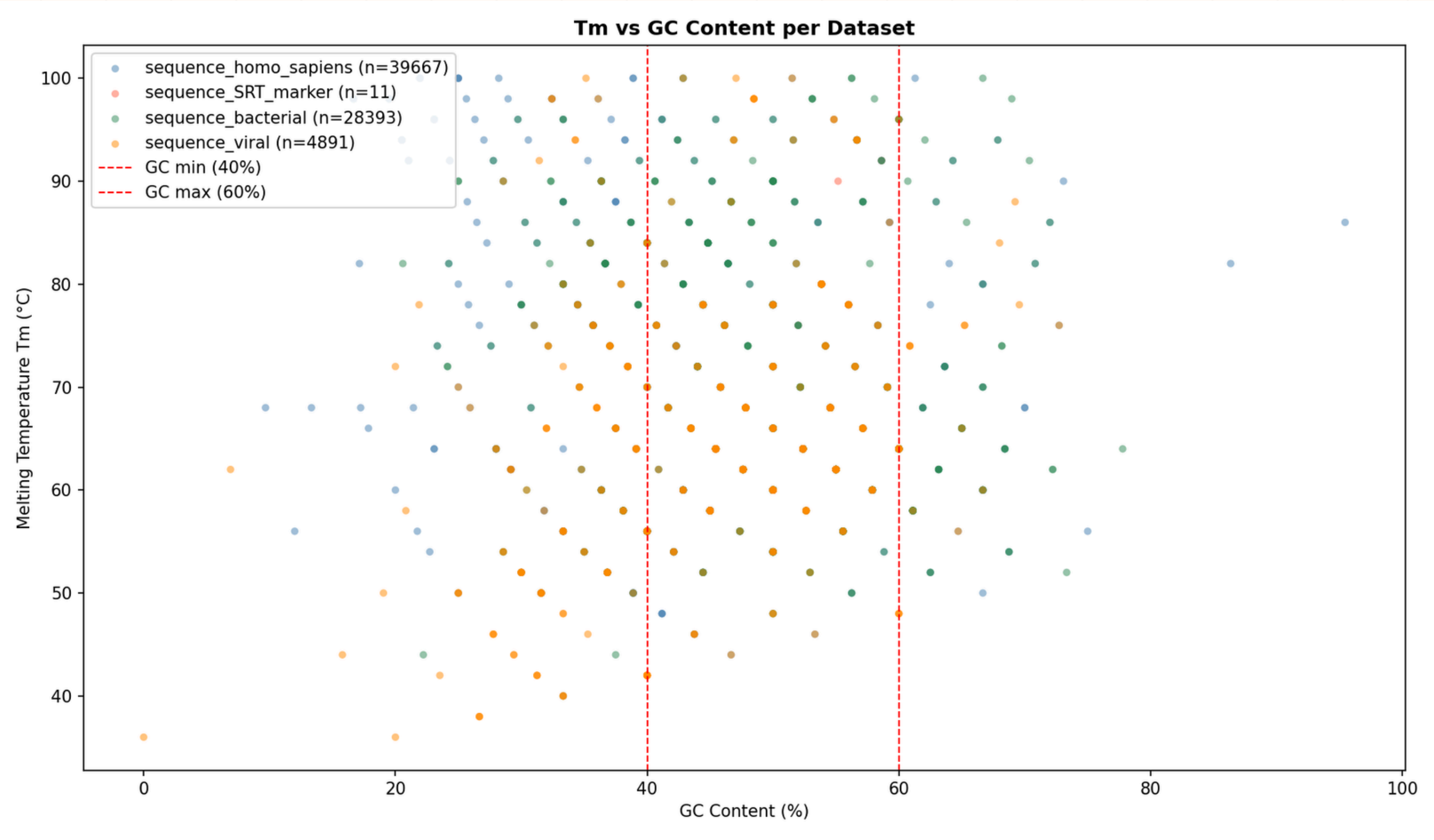
PCRpal is a Python CLI tool that takes a CSV or FASTA file with primer sequences and returns a full validation report including:

- Melting temperature (T_m)
- GC content
- Sequence length validation
- Self-complementarity check
- Primer dimer detection
- CSV report generation
- Multi-dataset comparison with plots

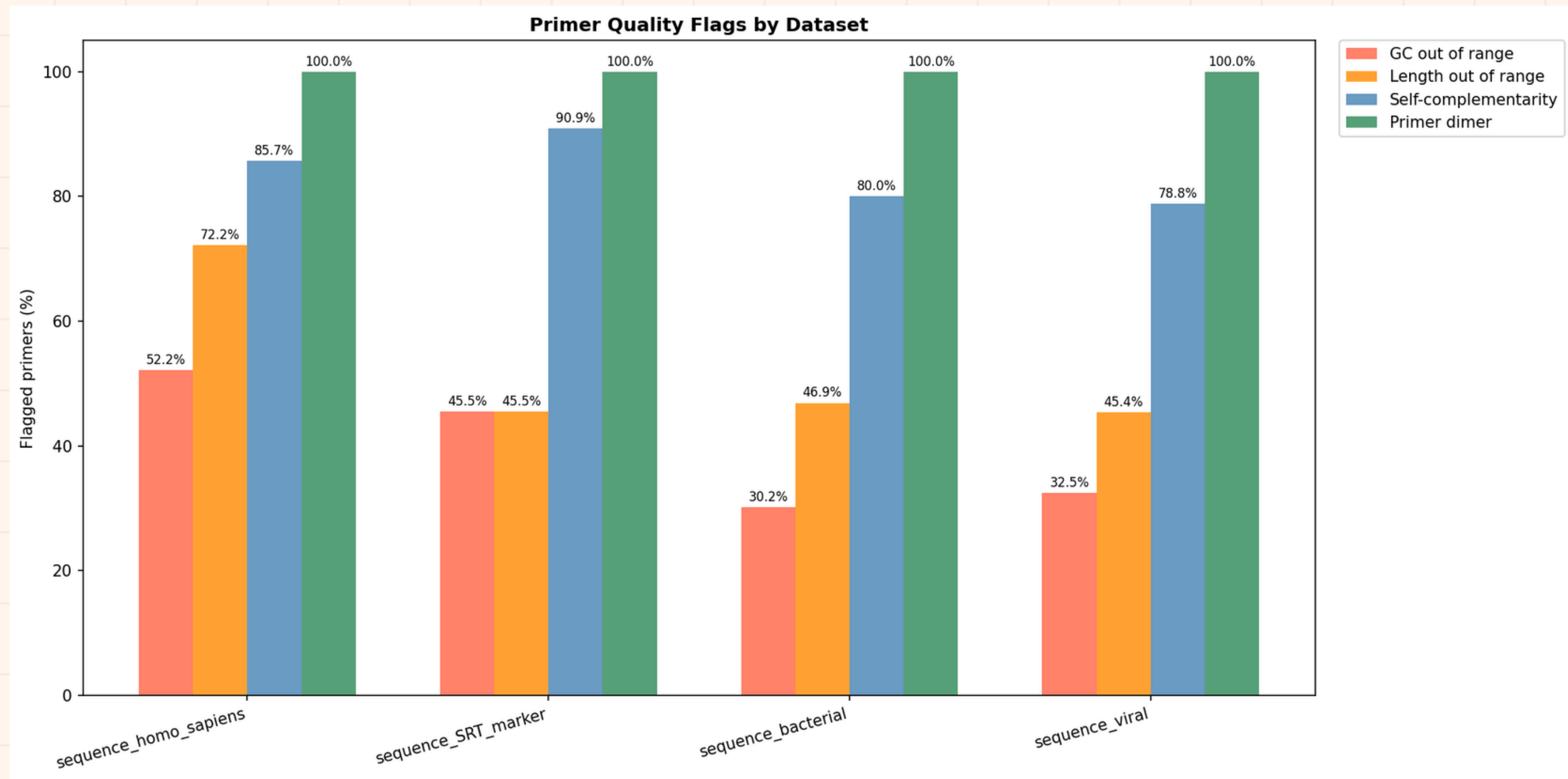
PCRpal consists of:

- **main.py** → orchestrates the pipeline, CLI interface
- **parser.py** → reads input file, validates sequences
- **calc.py** → calculates Tm, GC content, flags out-of-range values
- **dimers.py** → checks self-complementarity and primer dimers
- **report.py** → generates CSV report and plots

RESULTS



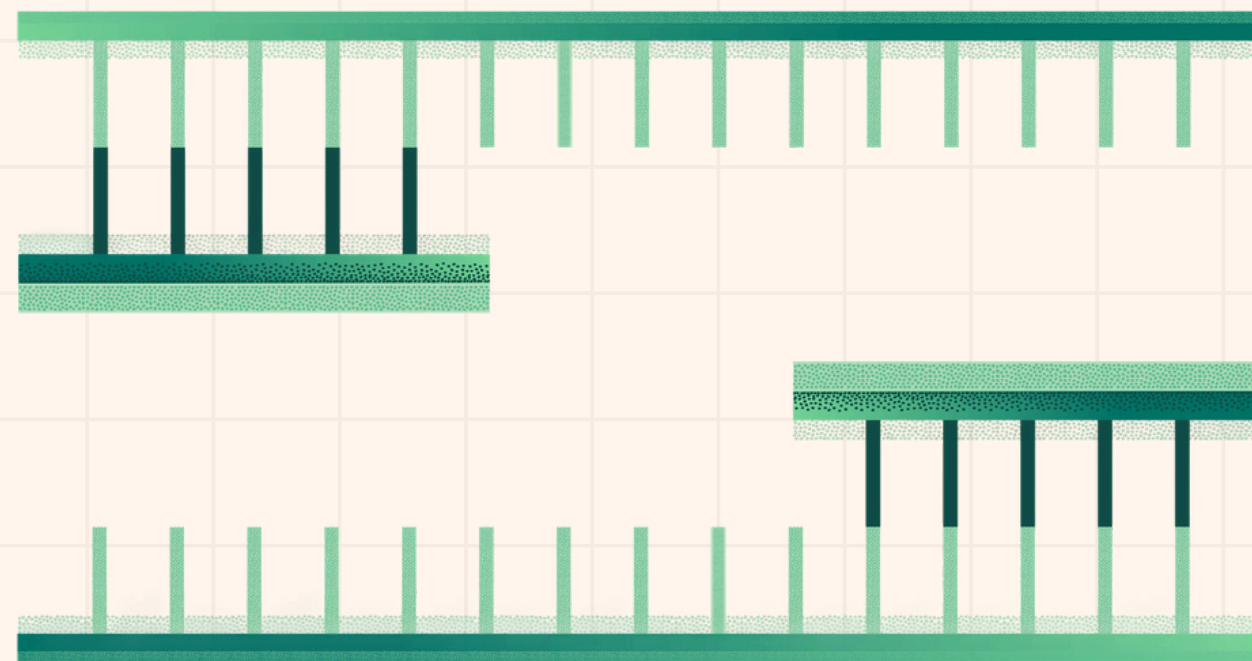
RESULTS



- **Bacterial dataset:** lowest GC out-of-range rate (30.2%)
- **Homo sapiens:** highest proportion of length outliers (72.2%)
- Primer dimer risk high across all datasets (79–100%)

FUTURE PLANS

- Integration with Biopython Bio.SeqUtils.MeltingTemp for more accurate T_m calculations
- Support for target sequence input and binding site visualization
- BLAST integration for off-target binding check
- GUI via Streamlit



BIBLIOGRAPHY

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<https://doi.org/10.1093/nar/25.16.3235>
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Thank you for your attention!